Lumbar Microdiskectomy

NeuroSynth - Automated Knowledge Synthesis

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Abstract

Abstract

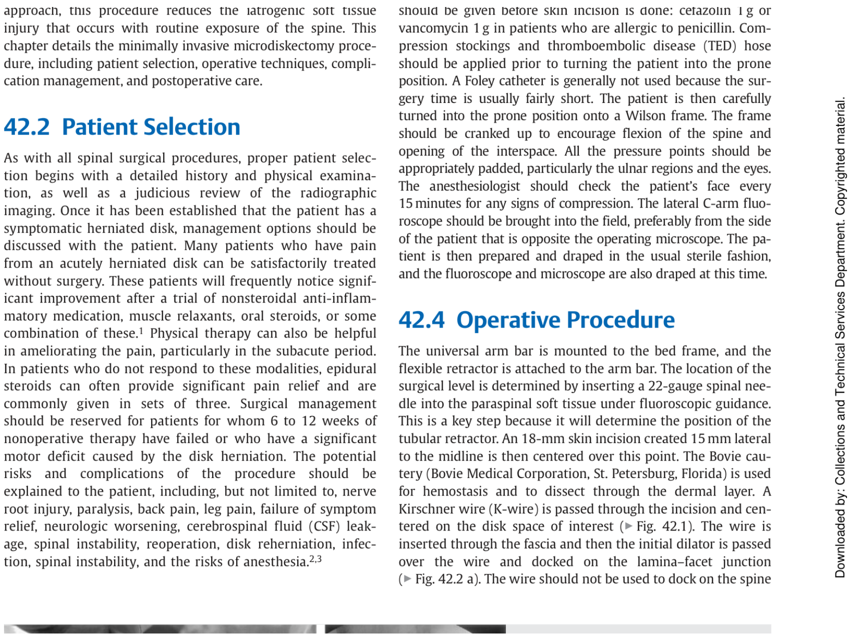
Background: Minimally invasive lumbar microdiskectomy has emerged as a cornerstone procedure in modern spine surgery, utilizing muscle-dilating techniques and specialized tubular retractors to reduce tissue trauma while maintaining therapeutic efficacy. This approach addresses symptomatic lumbar disk herniation through a significantly smaller incision compared to traditional open procedures, offering reduced morbidity and faster recovery times.

Key Points: Lumbar disk herniation represents one of the most common spinal pathologies requiring neurosurgical intervention. Conservative management including anti-inflammatory medications, physical therapy, and epidural steroid injections remains first-line treatment, with surgical intervention indicated after 6-12 weeks of failed conservative therapy or when significant motor deficits develop. Patient selection requires careful evaluation, with ideal candidates having focal neural foraminal stenosis correlating with clinical symptoms. Contraindications include large herniations causing central canal stenosis and cases requiring fusion procedures. The Maine Lumbar Spine Study Group demonstrated superior one-year outcomes in surgical patients (71% pain relief) compared to nonoperative management (43% relief), though Weber's landmark study showed convergent outcomes at ten years (55% versus 57% good results). Notably, clinical success appears unrelated to disk size, canal compromise degree, or motor weakness severity. Timing significantly influences outcomes, with patients treated within two months of symptom onset achieving superior results compared to those with prolonged conservative management.

Conclusions: Minimally invasive lumbar microdiskectomy provides effective treatment for appropriately selected patients with symptomatic disk herniation, offering earlier symptomatic relief and improved short-term functional outcomes compared to conservative management. While long-term outcomes may converge between treatment modalities, the procedure's reduced morbidity profile and faster recovery make it an attractive option for patients failing conservative therapy.

Keywords: minimally invasive lumbar diskectomy, lumbar disk herniation, radicular pain, foraminal stenosis, sciatic pain, Lasègue sign, epidural steroid injections, paraspinal musculature, neural foraminal stenosis, cerebrospinal fluid leakage, spondylolisthesis, interbody fusion

1 Introduction



approach, this procedure reduces the iatrogenic soft tissue injury that occurs with routine exposure of the spine. This chapter details the minimally invasive microdiskectomy procedure, including patient selection, operative techniques, complication management, and postoperative care.

42.2 Patient Selection

As with all spinal surgical procedures, proper patient selection begins with a detailed history and physical examination, as well as a judicious review of the radiographic imaging. Once it has been established that the patient has a symptomatic herniated disk, management options should be discussed with the patient. Many patients who have pain from an acutely herniated disk can be satisfactorily treated without surgery. These patients will frequently notice significant improvement after a trial of nonsteroidal anti-inflammatory medication, muscle relaxants, oral steroids, or some combination of these.[1] Physical therapy can also be helpful in ameliorating the pain, particularly in the subacute period. In patients who do not respond to these modalities, epidural steroids can often provide significant pain relief and are commonly given in sets of three. Surgical management should be reserved for patients for whom 6 to 12 weeks of nonoperative therapy have failed or who have a significant motor deficit caused by the disk herniation. The potential risks and complications of the procedure should be explained to the patient, including, but not limited to, nerve root injury, paralysis, back pain, leg pain, failure of symptom relief, neurologic worsening, cerebrospinal fluid (CSF) leakage, spinal instability, reoperation, disk reherniation, infection, spinal instability, and the risks of anesthesia.[2,3]

should be given before skin incision is done: cefazolin 1 g or vancomycin 1 g in patients who are allergic to penicillin. Compression stockings and thromboembolic disease (TED) hose should be applied prior to turning the patient into the prone position. A Foley catheter is generally not used because the surgery time is usually fairly short. The patient is then carefully turned into the prone position onto a Wilson frame. The frame should be cranked up to encourage flexion of the spine and opening of the interspace. All the pressure points should be appropriately padded, particularly the ulnar regions and the eyes. The anesthesiologist should check the patient's face every 15 minutes for any signs of compression. The lateral C-arm fluoroscope should be brought into the field, preferably from the side of the patient that is opposite the operating microscope. The patient is then prepared and draped in the usual sterile fashion, and the fluoroscope and microscope are also draped at this time.

42.4 Operative Procedure

The universal arm bar is mounted to the bed frame, and the flexible retractor is attached to the arm bar. The location of the surgical level is determined by inserting a 22-gauge spinal needle into the paraspinal soft tissue under fluoroscopic guidance. This is a key step because it will determine the position of the tubular retractor. An 18-mm skin incision created 15 mm lateral to the midline is then centered over this point. The Bovie cautery (Bovie Medical Corporation, St. Petersburg, Florida) is used for hemostasis and to dissect through the dermal layer. A Kirschner wire (K-wire) is passed through the incision and centered on the disk space of interest (▶ Fig. 42.1). The wire is inserted through the fascia and then the initial dilator is passed over the wire and docked on the lamina–facet junction (▶ Fig. 42.2 a). The wire should not be used to dock on the spine

Figure 1: Figure 42.1: (a) After localizing the skin incision

Introduction

Minimally invasive lumbar diskectomy represents a significant advancement in the surgical management of lumbar disk herniation, offering patients the benefits of reduced tissue trauma while maintaining the efficacy of traditional open procedures. This technique has emerged as one of the most commonly performed minimally invasive spine procedures, fundamentally changing the approach to symptomatic disk herniation through the application of muscle-dilating techniques that minimize iatrogenic soft tissue injury.

The clinical importance of lumbar disk herniation cannot be overstated, as it represents one of the most frequent causes of radicular pain and neurological dysfunction in the adult population. Patients typically present with a constellation of symptoms including back pain, sciatic pain, positive Lasègue sign (pain with straight leg raise), and sensory deficits that may progress to motor weakness if left untreated. The condition often manifests as referable foraminal stenosis secondary to posterolateral disk herniation, creating significant functional impairment and reduced quality of life.

Conservative management remains the initial treatment approach for most patients proper tube position is radiographically confirmed, the tube is locked into position with the flexible retractor arm, and the dilators are removed (▶ Fig. 42.5). The remainder of the procedure can be performed using loupe magnification, the endoscope, or the operating microscope, and the surgical technique is identical for any of these methods.

cautery should be minimized to reduce the amount of postoperative epidural scarring. Injectable hemostatic agents generally provide excellent hemostasis. The microknife is used to incise the annulus, and the disk herniation is removed using a combination of a small pituitary forcep and a sucker using a hand-over-fist technique. Ideally, the disk herniation should be gently

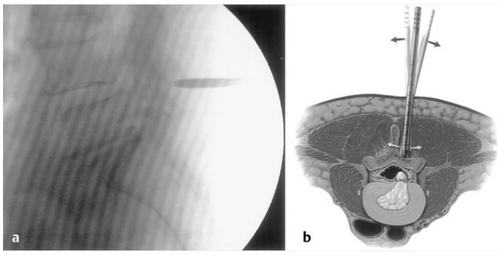


Fig. 42.2 **(a)** Fluoroscopic image demonstrating the initial dilator docked on the spine at the laminar–facet junction directly over the disk space. **(b)** The dilator is used in a sweeping motion to remove soft tissue away from the lamina and identify bony landmarks.

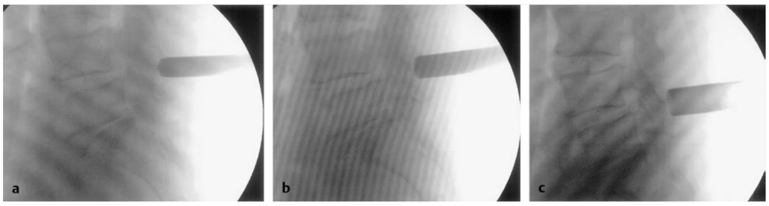


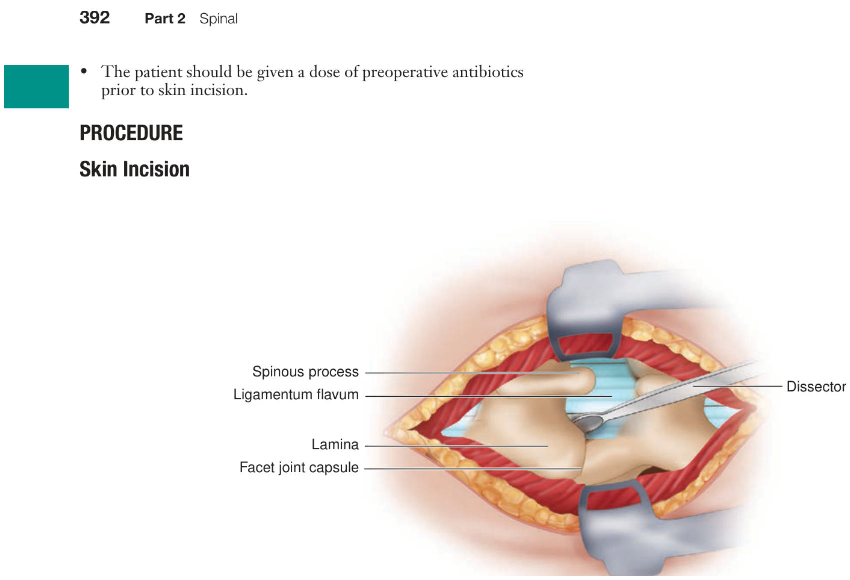
Figure 2: Figure 42.3: Serial fluoroscopic images demonstrating gradual dilation of muscle and fascia in preparation of tubular retractor placement. (a) A 14-mm

Minimally Invasive Lumbar Diskectomy

because it could inadvertently pass through the interlaminar space and cause a CSF leak or nerve root injury. Once the location of the dilator is confirmed with fluoroscopy, the K-wire is then removed from the field. Controlled forceful sweeping movements of the dilator in the cephalad–caudad and lateral–medial directions are then used to scrape the paraspinous musculature off the bony structures at the tip of the dilator (▶ Fig. 42.2 b). Incrementally larger dilators are sequentially passed over the initial dilator and advanced to the laminar–facet junction (▶ Fig. 42.3, ▶ Fig. 42.4). The numerical reading on the side of the final dilator is used to determine the appropriate tubular retractor length for the patient, ranging from 3 to 10 cm long. The diameter of the tube ranges from 14 to 26 mm, although 16- or 18-mm-diameter retractors are most commonly used for microdiskectomy. The appropriately sized tubular retractor is then passed over the final dilator. A variety of different medical device companies provide slightly different versions of the retractor; essentially, there is no functional difference between the various options. The tubular retractor should be docked on the lamina facet junction such that a small portion of the interlaminar space is visible inferiorly. Once the proper tube position is radiographically confirmed, the tube is locked into position with the flexible retractor arm, and the dilators are removed (▶ Fig. 42.5). The remainder of the procedure can be performed using loupe magnification, the endoscope, or the operating microscope, and the surgical technique is identical for any of these methods.

The microscope or endoscope is brought into the surgical field at this point and appropriately positioned and focused. The thin layer of soft tissue over the facet and lamina is removed using extended monopolar cautery. A straight or angled curet is used to expose the inferior edge of the lamina, repositioning the tube as needed to obtain proper exposure. A Kerrison punch (KMedic, Northvale, New Jersey) or a drill is then used to perform the laminotomy and medial facetectomy. Leaving the ligamentum flavum intact during the laminotomy decreases the probability of durotomy. Blunt dissection of the ligamentum flavum from the undersurface of the lamina and medial facet is safely and efficiently performed using an angled curet. The pedicle should be palpated (▶ Fig. 42.6) and serves as a marker for the extent of lateral bone removal. An angled curet or blunt nerve hook is passed rostrally and used to separate the origin of the ligamentum flavum from the underside of the lamina. The ligament is then separated from the dura and gently pulled inferiorly. The Kerrison is then used to remove the ligament in piecemeal fashion. The nerve root is gently retracted medially, exposing the disk space and any free disk fragments. Epidural veins are preserved as much as possible, and electro-cautery should be minimized to reduce the amount of postoperative epidural scarring. Injectable hemostatic agents generally provide excellent hemostasis. The microknife is used to incise the annulus, and the disk herniation is removed using a combination of a small pituitary forcep and a sucker using a hand-over-fist technique. Ideally, the disk herniation should be gently

Figure 3: Figure 42.2: (a) Fluoroscopic image demonstrating


392 **Part 2** Spinal

- The patient should be given a dose of preoperative antibiotics prior to skin incision.

PROCEDURE

Skin Incision

Figure 4: Figure 72.2: By palpating the anterior superior iliac crest, the L4-5 interspace can be localized for a rough

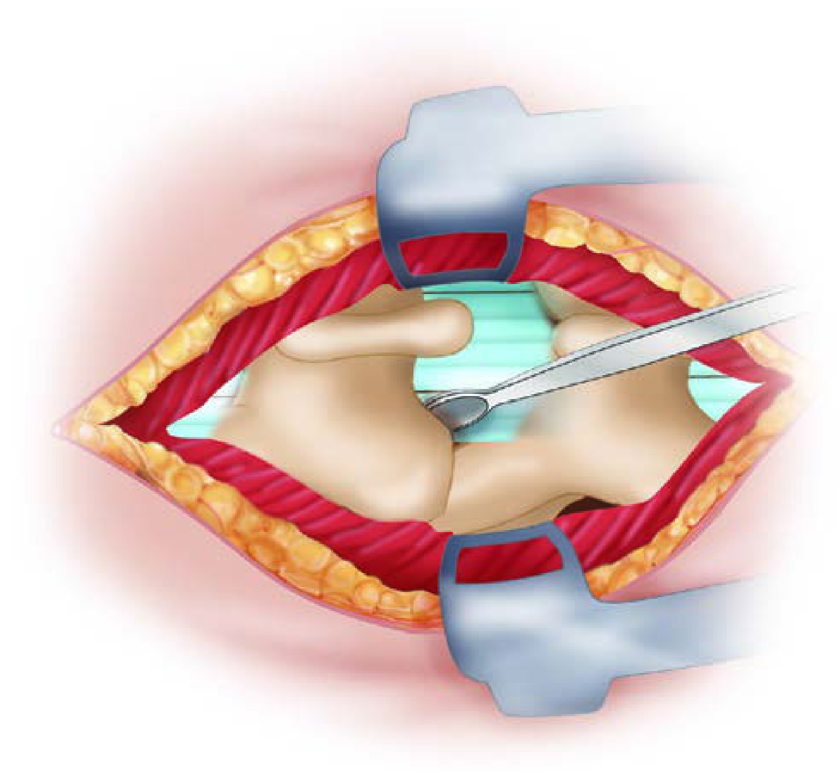


Figure 5: Figure 72.2: By palpating the anterior superior iliac crest, the L4-5 interspace can be localized for a rough estimation of the level. A spinal needle localization film helps determine the correct site for the exposure. After injection of local anesthetic with epinephrine into the skin and deeper tissues, a small incision just lateral to the midline should be made using a No. 10 blade

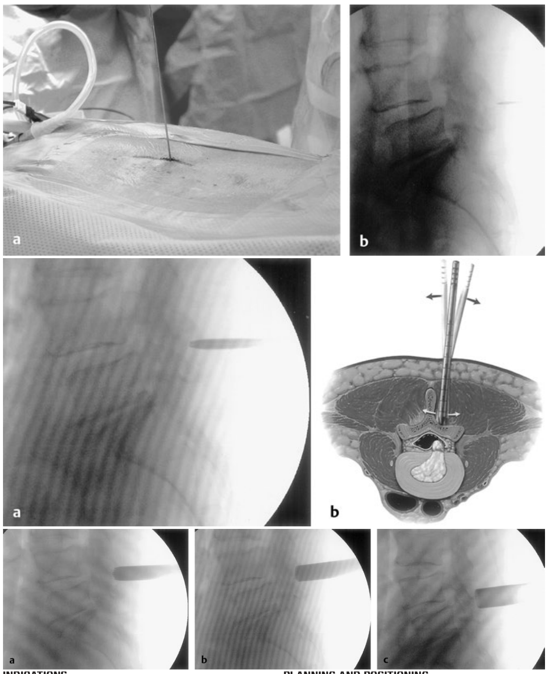
with acute disk herniation. A comprehensive nonoperative regimen typically includes nonsteroidal anti-inflammatory medications, muscle relaxants, oral steroids, and physical therapy, particularly during the subacute period. For patients who do not respond adequately to these initial measures, epidural steroid injections, commonly administered in series of three, can provide substantial pain relief. However, surgical intervention becomes necessary when conservative treatment fails after 6 to 12 weeks or when patients develop significant motor deficits caused by the disk herniation.

The evolution from traditional open diskectomy to minimally invasive techniques has been driven by the desire to reduce surgical morbidity while maintaining therapeutic effectiveness [FIGURE: 3da6af6e-bab]. The minimally invasive approach utilizes a muscle-dilating technique that preserves the integrity of the paraspinal musculature and reduces postoperative pain, leading to faster recovery times and improved patient satisfaction. This technique employs specialized tubular retractors and sequential dilators to create a working corridor to the spine, allowing surgeons to perform the same decompressive procedure through a significantly smaller incision [FIGURE: 22d1273e-8fc].

Patient selection for minimally invasive lumbar diskectomy requires careful consideration of multiple factors. Ideal candidates include those with symptomatic herniated disks who have failed conservative management, particularly when imaging studies demonstrate focal neural foraminal stenosis and disk herniation that correlates with clinical symptoms. However, certain contraindications must be recognized, including large disk herniations causing central canal stenosis, which may require more extensive laminectomy procedures, and cases involving recurrent disk herniation or concomitant instability from degenerative disease or spondylolisthesis, which may benefit from interbody fusion procedures.

The scope of minimally invasive lumbar diskectomy encompasses not only the technical aspects of the surgical procedure but also comprehensive patient evaluation, appropriate case selection, meticulous operative technique, and thorough postoperative management [FIGURE: bf46b63e-936]. Success depends on proper understanding of the anatomical considerations, mastery of the specialized instrumentation, and recognition of potential complications including nerve root injury, cerebrospinal fluid leakage, spinal instability, disk reherniation, and infection. The procedure can be performed using various visualization methods, including loupe magnification, endoscopy, or operating microscopy, with the surgical technique remaining fundamentally identical regardless of the chosen approach.

This comprehensive approach to minimally invasive lumbar diskectomy has established it as a cornerstone procedure in modern spine surgery, offering patients an effective treatment option with reduced morbidity compared to traditional open techniques while maintaining excellent clinical outcomes for appropriately selected cases.



INDICATIONS

- Patients with back pain, sciatic pain, Lasègue sign (pain with straight leg raise), or sensory deficit that fails to improve with conservative measures.
- Referable foraminal stenosis from posterolateral disk herniation.
- New or progressive motor deficits require more urgent surgical intervention.

CONTRAINDICATIONS

- Large disk herniation causing central canal stenosis requires a laminectomy.
- A more extensive procedure, such as interbody fusion, may be beneficial to patients with recurrent disk herniation or concomitant instability from degenerative disease or spondylolisthesis.

PLANNING AND POSITIONING

- Preoperative evaluation includes a thorough neurologic history and sensory, motor, and reflex examination. The sensory examination should be conducted to rule out any dermatomal distribution of sensory loss. Routine lateral, flexion, and extension lumbar spine films can show dynamic instability that may necessitate additional lumbar fusion. In the event of instability, a diskectomy would treat the radiculopathic leg pain but would not improve mechanical back pain. Magnetic resonance imaging (MRI) without contrast enhancement of the lumbar spine can show focal neural foraminal stenosis and disk herniation and allows selection of the appropriate level for diskectomy and foraminotomy.
- Either prone or lateral position is acceptable, depending on surgeon preference.

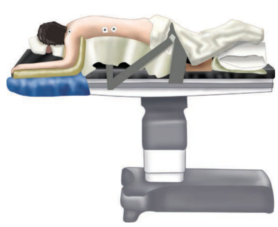


Figure 6: Introduction - Visuals

2 Epidemiology and Natural History

Epidemiology and Natural History

Incidence and Prevalence

Lumbar disc herniation represents one of the most common spinal pathologies encountered in neurosurgical practice, though precise epidemiological data vary considerably across studies due to differences in diagnostic criteria and population characteristics. The condition affects a substantial portion of the adult population, with clinical manifestations ranging from asymptomatic radiographic findings to debilitating radicular pain requiring surgical intervention.

The natural history of lumbar disc herniation has been extensively studied, though methodological limitations have complicated the interpretation of outcomes data. Weber's seminal prospective randomized study provides foundational insights into disease progression, demonstrating that operative and nonoperative treatment outcomes tend to converge over time. At one year follow-up, good results were reported in 33% of the nonoperative group versus 66% in the surgical group; however, by ten years, these outcomes had essentially equalized at 55% versus 57%, respectively. These findings suggest that while surgery provides earlier symptomatic relief, the long-term natural history may be similar regardless of initial treatment approach.

Demographics and Risk Factors

The demographic profile of patients with symptomatic lumbar disc herniation reveals several important patterns. The Maine Lumbar Spine Study Group, which followed more than 500 patients prospectively, demonstrated that surgically treated patients were initially more symptomatic at presentation compared to those managed nonoperatively, yet showed superior outcomes at one-year follow-up. Relief of back or leg pain was reported by 71% of operated patients compared with 43% of the nonoperative group, with high satisfaction levels and improved quality of life documented for the operative cohort.

Occupational factors appear to influence both disease incidence and outcomes. Alaranta and colleagues identified that patients with sciatica but no evidence of nerve root compression had a high incidence of physically strenuous jobs and demonstrated distinctly worse natural history compared to those with documented root compression. This subgroup also exhibited characteristics suggesting psychosocial and behavioral factors in symptom perpetuation, including generalized pain patterns and lower pain thresholds.

Disease Progression and Natural History

The natural history of lumbar disc herniation demonstrates considerable variability, with multiple factors influencing clinical outcomes. Bush and colleagues reported results from 159 patients with CT-confirmed disc herniations treated with epidural steroid injections, noting that 91% avoided surgery. However, this study highlights a fundamental challenge in natural history research, as the intervention itself may alter disease progression through biochemical mechanisms that are still being elucidated.

The Spine Patient Outcomes Research Trial (SPORT) investigation, designed as a rigorous randomized controlled study, encountered significant methodological challenges including high crossover rates between treatment groups. Despite these limitations, the study demonstrated improvement in bodily pain, physical function, and Oswestry Disability Index scores regardless of intervention. In an as-treated analysis, surgical treatment showed statistically superior results compared with nonoperative treatment, with these benefits proving durable at eight-year follow-up.

Factors Affecting Prognosis

Several clinical and radiographic factors have been identified as prognostic indicators in lumbar disc herniation. Importantly, clinical success following both operative and nonoperative treatment appears unrelated to disc size, percentage canal compromise, or degree of motor weakness. This finding challenges traditional assumptions about the relationship between anatomical severity and clinical outcomes.

Timing of intervention represents a critical prognostic factor. Rotheorl and colleagues demonstrated that patients with symptom duration exceeding two months had statistically significantly worse outcomes than those operated within two months of presentation. Similarly, Hurme and Alaranta found optimal results in patients operated within two months of disabling sciatica onset. Nygaard and colleagues reported inferior results in patients with leg pain persisting eight months or longer, while analysis of Swedish registry data including over 27,000 patients confirmed similar findings.

The influence of psychosocial factors on outcomes cannot be understated. Patients involved in litigation, disability claims, or workers' compensation cases demonstrate different outcome patterns, though the Maine study found no difference in return-to-work rates between operative and nonoperative groups at long-term follow-up. The presence of concurrent psychological issues, generalized pain syndromes, and lower pain thresholds appears to negatively influence prognosis regardless of treatment modality.

Long-term Outcomes

Long-term follow-up data provide valuable insights into disease progression. The Maine Lumbar Spine Study Group's ten-year results demonstrated that 85% of surgically treated patients and 82% of nonoperatively treated patients had available data for analysis. A significantly larger percentage of surgical patients reported relief of low back and leg pain compared to those treated nonoperatively, with superior functional outcomes and satisfaction scores. However, improvement in dominant symptoms was documented in both treatment groups, and work and disability status were similar between groups at final follow-up.

The durability of surgical outcomes is evidenced by reoperation rates, with 20% of operative patients requiring additional surgery and 16% of initially nonoperatively treated patients eventually undergoing operation. These crossover rates underscore the dynamic nature of treatment decisions and the importance of individualized care approaches.

Understanding the epidemiology and natural history of lumbar disc herniation is essential for appropriate patient counseling and treatment planning. While surgery provides earlier symptomatic relief, the convergence of long-term outcomes between operative and nonoperative approaches emphasizes the importance of careful patient selection and realistic expectation setting. The multifactorial nature of prognosis, incorporating anatomical, temporal, occupational, and psychosocial elements, necessitates comprehensive evaluation in clinical decision-making.

3 Clinical Presentation and Diagnosis

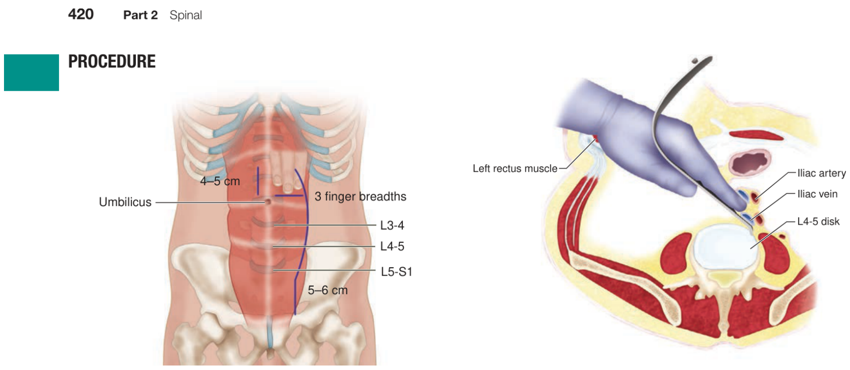


Figure 7: Figure 78.5: At the L4-5 level, mobilization of the left iliac vessels from left

Clinical Presentation and Diagnosis

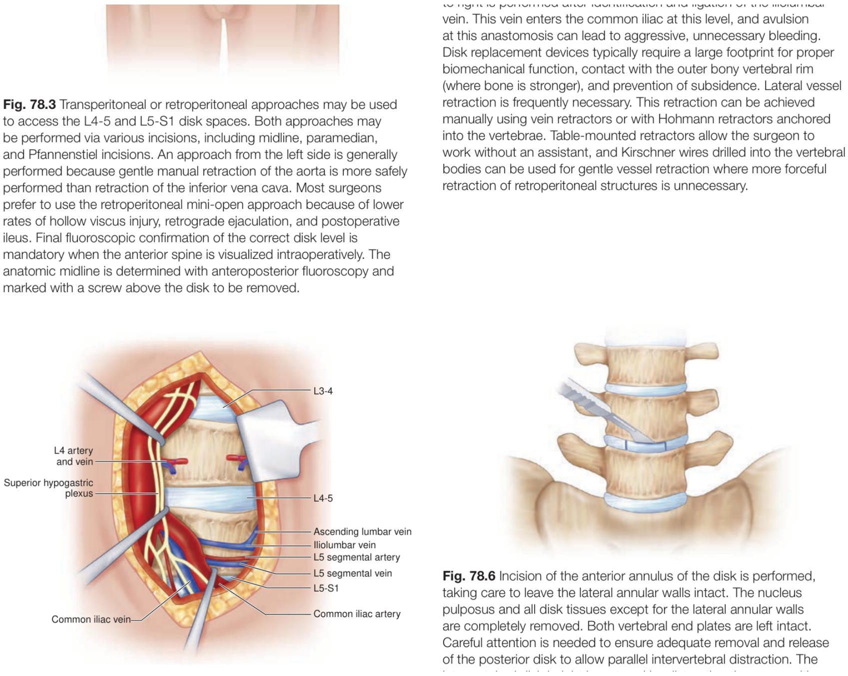
Clinical Presentation

The clinical presentation of lumbar disc disease encompasses a spectrum of symptoms ranging from localized back pain to radiculopathy with neurologic deficits. The pathophysiology underlying these symptoms involves both mechanical compression and inflammatory processes that contribute to pain generation and neural dysfunction.

Discogenic Pain

Discogenic pain typically manifests as axial low back pain that may be accompanied by referred pain patterns. The intervertebral disc itself contains nociceptive nerve fibers, particularly in the outer layers of the annulus fibrosus, which can generate pain when stimulated by mechanical or chemical irritants.^153,154^ Studies have demonstrated that painful discs exhibit increased innervation compared to normal discs, with nerve growth factor-dependent neurons contributing to inflammatory pain responses.^158^ The pain is often described as deep, aching, and may worsen with activities that increase intradiscal pressure, such as sitting, forward flexion, or coughing.

Radicular Symptoms



vein. This vein enters the common iliac at this level, and avulsion at this anastomosis can lead to aggressive, unnecessary bleeding. Disk replacement devices typically require a large footprint for proper biomechanical function, contact with the outer bony vertebral rim (where bone is stronger), and prevention of subsidence. Lateral vessel retraction is frequently necessary. This retraction can be achieved manually using vein retractors or with Hohmann retractors anchored into the vertebrae. Table-mounted retractors allow the surgeon to work without an assistant, and Kirschner wires drilled into the vertebral bodies can be used for gentle vessel retraction where more forceful retraction of retroperitoneal structures is unnecessary.

Fig. 78.3 Transperitoneal or retroperitoneal approaches may be used to access the L4-5 and L5-S1 disk spaces. Both approaches may be performed via various incisions, including midline, paramedian, and Pfannenstiel incisions. An approach from the left side is generally performed because gentle manual retraction of the aorta is more safely performed than retraction of the inferior vena cava. Most surgeons prefer to use the retroperitoneal mini-open approach because of lower rates of hollow viscus injury, retrograde ejaculation, and postoperative ileus. Final fluoroscopic confirmation of the correct disk level is mandatory when the anterior spine is visualized intraoperatively. The anatomic midline is determined with anteroposterior fluoroscopy and marked with a screw above the disk to be removed.

Fig. 78.6 Incision of the anterior annulus of the disk is performed, taking care to leave the lateral annular walls intact. The nucleus pulposus and all disk tissues except for the lateral annular walls are completely removed. Both vertebral end plates are left intact. Careful attention is needed to ensure adequate removal and release of the posterior disk to allow parallel intervertebral distraction. The

Figure 8: Figure 78.4: Prevertebral tissue at the L5-S1 level is opened using a vertical

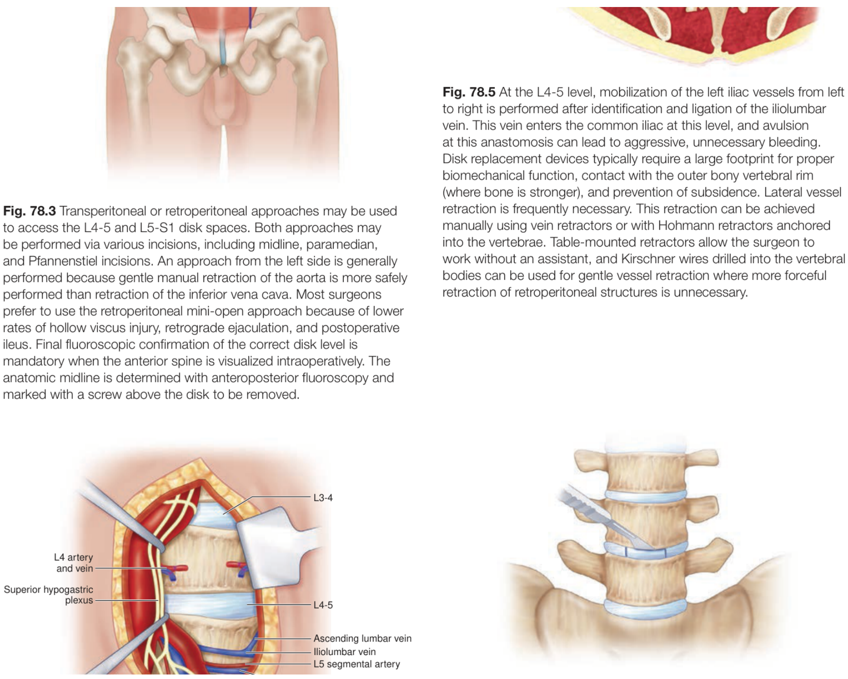


Fig. 78.5 At the L4-5 level, mobilization of the left iliac vessels from left to right is performed after identification and ligation of the iliolumbar vein. This vein enters the common iliac at this level, and avulsion at this anastomosis can lead to aggressive, unnecessary bleeding. Disk replacement devices typically require a large footprint for proper biomechanical function, contact with the outer bony vertebral rim (where bone is stronger), and prevention of subsidence. Lateral vessel retraction is frequently necessary. This retraction can be achieved manually using vein retractors or with Hohmann retractors anchored into the vertebrae. Table-mounted retractors allow the surgeon to work without an assistant, and Kirschner wires drilled into the vertebral bodies can be used for gentle vessel retraction where more forceful retraction of retroperitoneal structures is unnecessary.

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Figure 9: Figure 78.6: Incision of the anterior annulus of the disk is performed,

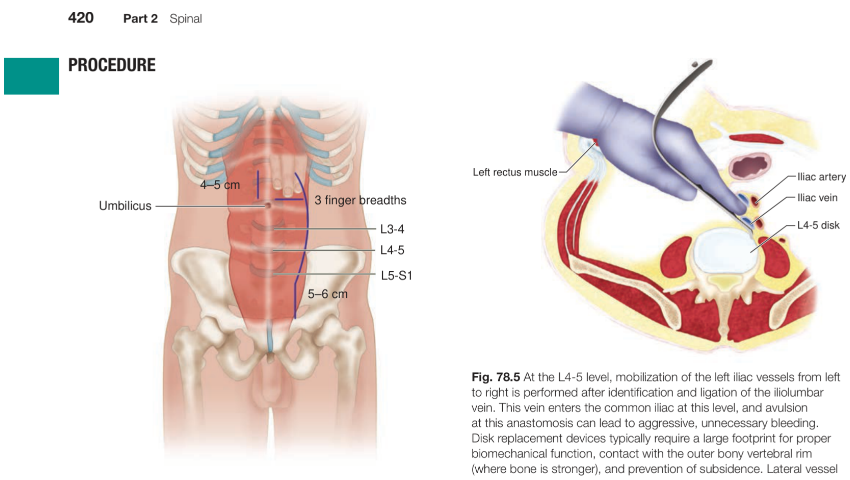


Figure 10: Figure 78.3: Transperitoneal or retroperitoneal approaches may be used

When disc herniation results in nerve root compression or irritation, patients typically develop radicular symptoms. The classic presentation includes leg pain that follows a dermatomal distribution, often described as sharp, burning, or electric-like in quality. The pain frequently radiates below the knee and may be accompanied by numbness, tingling, or weakness in the affected nerve root distribution. Patients often report that leg pain is more severe than back pain, which is characteristic of radiculopathy.

The straight leg raise (SLR) test remains a cornerstone of physical examination for lumbar radiculopathy. A positive test reproduces the patient's radicular symptoms when the affected leg is raised between 30 and 70 degrees while the patient is supine. The contralateral SLR test, which reproduces symptoms in the affected leg when the unaffected leg is raised, is less sensitive but more specific for disc herniation.

Neurologic Deficits

Motor weakness may develop in the distribution of the compressed nerve root. Common patterns include weakness of ankle dorsiflexion and great toe extension with L5 radiculopathy, and weakness of ankle plantarflexion with S1 radiculopathy. Sensory deficits typically follow dermatomal patterns, with decreased sensation in the first web space for L5 involvement and the lateral foot for S1 involvement. Reflex changes include diminished or absent patellar reflex with L4 radiculopathy and Achilles reflex with S1 radiculopathy.

Cauda Equina Syndrome

Cauda equina syndrome represents a surgical emergency that may occur with large central disc herniations. The presentation includes bilateral lower extremity weakness, saddle anesthesia, bowel and bladder dysfunction, and severe back pain. Progressive neurologic deficit, particularly in the setting of cauda equina syndrome, constitutes an absolute indication for urgent surgical intervention.^72^

Inflammatory Mechanisms

Recent research has elucidated the significant role of inflammatory mediators in disc-related pain. Herniated disc material contains high levels of inflammatory substances, including phospholipase A2, prostaglandin E2, and various cytokines.^145,147,149^ Tumor necrosis factor-alpha (TNF-) and interleukin-1 have been identified as key inflammatory mediators that contribute to both pain generation and neural dysfunction.^97,98^ These inflammatory substances can cause direct neural irritation even without mechanical compression, explaining why some patients experience significant symptoms despite minimal anatomic compression on imaging studies.

The inflammatory response also involves the release of matrix metalloproteinases and other degradative enzymes that can further compromise neural function.^98,99^ High mobility group box 1 (HMGB1), released by necrotic cells, has been shown to trigger inflammatory responses and contribute to pain generation in animal models.^125^ Understanding these inflammatory pathways has led to targeted therapeutic approaches, including epidural steroid injections and experimental anti-TNF- treatments.

Diagnostic Workup

Clinical History and Physical Examination

A comprehensive history should focus on the onset, duration, and character of symptoms, as well as aggravating and relieving factors. The presence of neurologic symptoms, including weakness, numbness, or bowel and bladder dysfunction, must be carefully assessed. Physical examination should include assessment of lumbar range of motion, neurologic testing of the lower extremities, and provocative maneuvers such as the straight leg raise test.

The clinical presentation should be correlated with imaging findings to establish a diagnosis. A patient presenting with left L4-L5 disc herniation should ideally demonstrate leg pain radiating to the dorsum of the foot, weakness of toe dorsiflexion, decreased sensation in the first dorsal web space, and positive straight leg raise testing.^72^ However, many patients present with less classic findings, and approximately 85% success rates have been reported even in patient groups with fewer objective motor and sensory findings.^72^

Imaging Studies

Magnetic resonance imaging (MRI) has become the gold standard for evaluating lumbar disc pathology. MRI can demonstrate disc herniation, nerve root compression, and associated findings such as Modic changes in adjacent vertebrae.^108^ The presence of vertebral endplate signal changes (Modic changes) has been associated with discogenic pain and may indicate ongoing inflammatory processes.^109^

Plain radiographs, while less sensitive for soft tissue pathology, can provide valuable information about spinal alignment, disc space height, and the presence of degenerative changes. Computed tomography (CT) may be useful when MRI is contraindicated or when bony detail is required.

Advanced Diagnostic Techniques

Provocative discography remains controversial but may be considered in select cases where the diagnosis is unclear. This procedure involves injection of contrast material into the disc space to reproduce the patient's symptoms and assess disc morphology.^92^ However, the clinical utility of discography has been questioned due to concerns about false-positive results and potential complications.^162,163^

Selective nerve root injections can serve both diagnostic and therapeutic purposes. These injections can help identify the symptomatic level in patients with multilevel pathology and provide temporary symptom relief.^66^ The diagnostic value is particularly useful when clinical and imaging findings are discordant.

Laboratory Studies

Laboratory studies are generally not required for routine disc disease evaluation. However, inflammatory markers such as erythrocyte sedimentation rate and C-reactive protein may be elevated in cases of discitis or other infectious processes.^106^ Recent research has investigated the potential role of bacterial infection in some cases of disc degeneration, though this remains an area of ongoing investigation.^110,111^

Differential Diagnosis

The differential diagnosis of lumbar disc disease includes other causes of low back pain and radiculopathy. Spinal stenosis, facet arthropathy, piriformis syndrome, and hip pathology should be considered. Systemic conditions such as diabetes mellitus may contribute to disc degeneration and should be evaluated when clinically indicated.^131,132^ The presence of red flag symptoms, including fever, unexplained weight loss, or progressive neurologic deficits, should prompt evaluation for more serious conditions such as malignancy or infection.

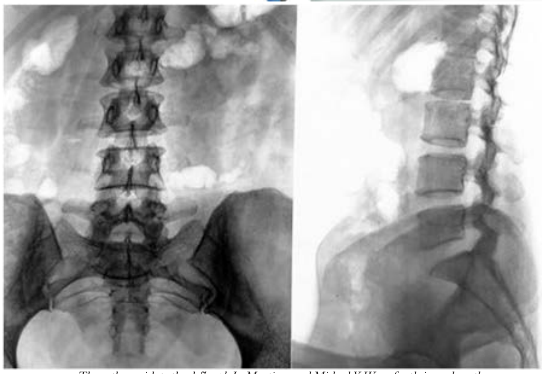
The diagnosis of lumbar disc disease requires careful integration of clinical presentation, physical examination findings, and appropriate imaging studies. Understanding the inflammatory mechanisms underlying disc-related pain has enhanced our ability to provide targeted treatments and improve patient outcomes. Early recognition of progressive neurologic deficits or cauda equina syndrome is crucial for optimal management and prevention of permanent neurologic injury.

4 Surgical Technique

Surgical Technique: Minimally Invasive Lumbar Discectomy

Patient Positioning and Setup

Position the patient prone on a Wilson frame under general anesthesia. **Anatomical Why**: The prone position provides optimal access to the posterior spinal elements while the Wilson frame flexion opens the interlaminar space by increasing the distance between adjacent laminae. **Safety Why**: Proper positioning with adequate padding of pressure



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INDICATIONS

- Lumbar disk arthroplasty (LDA) is indicated as a treatment of chronic, incapacitating low back pain that is diskogenic in origin at the L4-5 or L5-S1 level and not accompanied by neural element compression resulting in claudication or radiculopathy. Diagnosis should be documented with magnetic resonance imaging (MRI), plain lumbar spine x-rays, and positive results of provocative diskography of the pathologic level.
- To be considered for surgery, patients should have failed at least 6 months of conservative nonsurgical therapies and ideally be 18 to 50 years old. Patients who have previously undergone posterior disk interventions, such as diskectomy or nucleolysis, may be candidates for LDA as long as no acute neural compression is present and the remaining facet anatomy is sufficient to prevent distraction of the disk space and provide stability of the segment to be operated.

CONTRAINDICATIONS

- Assuming a patient's general medical condition is adequate to undergo elective spine surgery, contraindications to this procedure are active diskitis, previous (failed) fusion surgery at the pathologic level, malignancy, fracture or spondylolysis of the adjacent vertebrae, spondylolisthesis of the pathologic segment, osteopenia insufficient to support the disk prosthesis, and advanced facet arthrosis.
- As with other anterior lumbar spine procedures, relative contraindications to this approach include the presence of an infrarenal aortic aneurysm, congenital or iatrogenic genitourinary anatomic abnormalities such as an ipsilateral single ureter or kidney, or a history of previous retroperitoneal surgery.

PLANNING AND POSITIONING

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or kidney, or a history of previous retroperitoneal surgery.

PLANNING AND POSITIONING

Fig. 78.1 The patient is positioned supine on the operating table. The lumbar spine is placed in extension; the patient's legs are abducted if a lithotomy position is used. The surgeon performs the procedure standing between the legs of the patient if the lithotomy position is used. This position gives the surgeon a slightly more centered approach during prosthesis placement, which can be crucial to the success of the device.

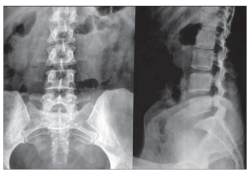

Figure 11: Clinical Presentation and Diagnosis - Visuals

points, particularly the ulnar regions and eyes, prevents positioning-related complications. **Outcome Why**: Frame flexion improves visualization of the disc space and reduces the need for extensive bone removal.

Apply compression stockings and thromboembolic disease (TED) hose prior to positioning to prevent venous stasis. Crank up the Wilson frame to encourage spinal flexion and interspace opening. Pad all pressure points appropriately, with particular attention to the ulnar regions and eyes. Instruct the anesthesiologist to check the patient's face every 15 minutes for signs of compression.

[IMAGE: Overview - Patient positioned prone on Wilson frame with padding and monitoring setup]

Position the lateral C-arm fluoroscope from the side opposite the operating microscope to avoid interference during the procedure. Prepare and drape the patient in sterile fashion, including sterile draping of the fluoroscope and microscope.

Incision Planning and Initial Access

Mount the universal arm bar to the bed frame and attach the flexible retractor to the arm bar. **Anatomical Why**: The retractor system provides stable, hands-free exposure while minimizing soft tissue trauma through controlled retraction.

Determine the surgical level by inserting a 22-gauge spinal needle into the paraspinal soft tissue under fluoroscopic guidance. **Safety Why**: Accurate level localization prevents wrong-level surgery, a serious complication that occurs in up to 4% of spine procedures. **Outcome Why**: Precise targeting reduces operative time and tissue dissection.

Create an 18-mm skin incision centered 15 mm lateral to the midline over the localized point. **Anatomical Why**: This lateral approach avoids the interspinous ligament and provides a more direct trajectory to the lamina-facet junction. Use Bovie cautery for hemostasis and dissection through the dermal layer.

[IMAGE: Detail - Fluoroscopic image showing needle localization and incision placement]

Sequential Dilation and Retractor Placement

Pass a Kirschner wire (K-wire) through the incision and center it on the disc space of interest [FIGURE: 42.1]. **Safety Why**: The K-wire serves as a guide for subsequent dilators but should not be used to dock directly on the spine as it could inadvertently pass through the interlaminar space causing CSF leak or nerve root injury.

Insert the wire through the fascia and pass the initial dilator over the wire, docking it on the lamina-facet junction [FIGURE: 42.2]. **Anatomical Why**: The lamina-facet junction provides a stable bony landmark that is consistently located lateral to the neural elements. Confirm proper dilator location with fluoroscopy before removing the K-wire.

Perform controlled forceful sweeping movements of the dilator in cephalad-caudad and lateral-medial directions to scrape the paraspinous musculature off the bony structures. **Outcome Why**: This technique preserves muscle fiber integrity better than sharp dissection, reducing postoperative pain and facilitating faster recovery.

[IMAGE: Intraop - Sequential dilator placement showing progressive muscle dilation]

Pass incrementally larger dilators sequentially over the initial dilator, advancing each to the laminar-facet junction [FIGURE: 42.3, 42.4]. **Safety Why**: Sequential dilation minimizes muscle trauma compared to direct insertion of large retractors. Use the numerical reading on the final dilator to determine the appropriate tubular retractor length (3-10 cm) for the patient's anatomy.

Select a 16- or 18-mm diameter tubular retractor for most microdiscectomy procedures. **Anatomical Why**: These diameters provide adequate visualization while minimizing soft tissue disruption. Pass the appropriately sized tubular retractor over the final dilator and dock it on the lamina-facet junction such that a small portion of the interlaminar space is visible inferiorly.

Confirm proper tube position radiographically, then lock the tube into position with the flexible retractor arm and remove all dilators [FIGURE: 42.5]. **Safety Why**: Radiographic confirmation prevents malposition that could lead to inadequate exposure or neural injury.

[IMAGE: Schematic - Cross-sectional view showing retractor position relative to neural elements]

Microscopic Exposure and Bone Work

Bring the operating microscope into the surgical field and position it appropriately. **Outcome Why**: Microscopic visualization provides superior illumination and magnification compared to loupe magnification, improving precision and reducing complications.

Remove the thin layer of soft tissue over the facet and lamina using extended monopolar cautery. Use a straight or angled curet to expose the inferior edge of the lamina, repositioning the tube as needed for optimal exposure. **Anatomical Why**: The inferior laminar edge serves as the superior boundary for the laminotomy.

Perform laminotomy and medial facetectomy using a Kerrison punch or drill. **Safety Why**: Leave the ligamentum flavum intact during bone removal to decrease the probability of durotomy. **Anatomical Why**: The ligamentum flavum provides a protective barrier between the bone work and the underlying dura.

Palpate the pedicle to serve as a marker for the extent of lateral bone removal [FIGURE: 42.6]. **Safety Why**: The pedicle represents the lateral limit of safe bone removal; extending beyond this landmark risks injury to the exiting nerve root.

[IMAGE: Detail - Microscopic view showing anatomical landmarks during bone removal]

Ligamentum Flavum Removal and Neural Exposure

Perform blunt dissection of the ligamentum flavum from the undersurface of the lamina and medial facet using an angled curet. **Safety Why**: Blunt dissection minimizes risk of dural tear compared to sharp dissection techniques.

Pass an angled curet or blunt nerve hook rostrally to separate the origin of the ligamentum flavum from the underside of the lamina. **Anatomical Why**: The ligamentum flavum has firm attachments to the lamina that must be carefully released to allow safe removal.

Separate the ligament from the dura and gently pull it inferiorly. Use the Kerrison punch to remove the ligament in piecemeal fashion. **Safety Why**: Piecemeal removal allows controlled visualization of underlying structures and reduces risk of inadvertent dural injury.

[IMAGE: Intraop - Step-by-step ligamentum flavum removal technique]

Disc Herniation Identification and Removal

Gently retract the nerve root medially to expose the disc space and any free disc fragments. **Safety Why**: Gentle retraction prevents nerve root injury while providing adequate visualization. **Outcome Why**: Proper nerve root retraction is essential for complete herniation removal and symptom relief.

Preserve epidural veins as much as possible and minimize electrocautery use to reduce postoperative epidural scarring. **Outcome Why**: Excessive scarring can lead to persistent pain and complicate revision surgery if needed. Use injectable hemostatic agents for excellent hemostasis when needed.

Incise the annulus using a microknife. **Anatomical Why**: A controlled annular incision allows access to the disc space while minimizing damage to remaining annular fibers. Remove the disc herniation using a combination of small pituitary forceps and suction with a hand-over-fist technique.

Safety Why: The hand-over-fist technique prevents inadvertent advancement of instruments toward neural structures. **Outcome Why**: Complete removal of loose fragments reduces the risk of recurrent herniation while preserving disc height and spinal stability.

[IMAGE: Detail - Disc herniation removal technique showing instrument positioning]

Closure and Final Considerations

Gently remove the disc herniation, ideally as a single piece when possible. **Outcome Why**: En bloc removal reduces the likelihood of retained fragments that could cause persistent symptoms or recurrent herniation.

Irrigate the surgical site and achieve hemostasis using injectable hemostatic agents rather than extensive electrocautery. **Safety Why**: Thorough irrigation removes debris and blood clots that could contribute to scarring or infection.

Remove the tubular retractor system under direct visualization to ensure no neural structures are trapped. **Safety Why**: Direct visualization during retractor removal prevents inadvertent neural injury that could occur with blind removal.

The minimally invasive approach reduces iatrogenic soft tissue injury compared to traditional open techniques while maintaining the same principles of neural decompression and disc removal. **Outcome Why**: Reduced tissue trauma translates to decreased postoperative pain, shorter hospital stays, and faster return to normal activities compared to open discectomy techniques.

[IMAGE: Overview - Final surgical site appearance after retractor removal]

This technique provides effective treatment for symptomatic disc herniation while minimizing surgical morbidity through the muscle-dilating approach and preservation of normal anatomical structures.

5 Outcomes and Prognosis

Outcomes and Prognosis

The long-term outcomes and prognosis for patients with lumbar disc disease vary significantly depending on the specific pathology, treatment approach, and individual patient factors. Understanding these outcomes is crucial for both clinicians and patients when making treatment decisions and setting realistic expectations for recovery.

Natural History and Conservative Treatment Outcomes

The natural history of lumbar disc disease demonstrates considerable variability, with many patients experiencing improvement over time without surgical intervention. In a prospective series by Bush and colleagues, 91% of patients with CT-confirmed disc herniations treated with epidural steroid injections avoided surgery, though this finding must be interpreted cautiously as it represents an interventional rather than truly natural history study.60 The effectiveness of conservative management has been further supported by analysis of SPORT data, which revealed that patients receiving epidural steroid injections were more likely to switch from surgical to nonsurgical treatment groups.61

Pharmacologic interventions, particularly nonsteroidal antiinflammatory drugs (NSAIDs), remain first-line treatments for disc-related symptoms. The rationale for their use stems from the understanding that local and systemic inflammatory reactions participate significantly in pain generation.58 Short-term narcotic use may be beneficial in acute settings, though extended use should be avoided, with courses preferably limited to 2-3 days. Oral corticosteroids have shown modest benefits, with Goldberg et al. demonstrating statistically significant functional improvement in patients with acute sciatica treated with a 15-day tapered dose of oral prednisone, though pain improvement was not observed.58

Selective transforaminal steroid injections have emerged as an effective intermediate treatment option. Wang and colleagues studied 69 patients who had failed noninvasive care and requested surgery; instead, these patients underwent transforaminal steroid injections at the affected root level. Remarkably, 77% achieved clinical resolution and had not undergone surgery at an average follow-up of 1.5 years.66 Clinical success was notably independent of disc size, percentage canal compromise, or degree of motor weakness, suggesting that anatomical severity may not predict treatment response.

Recent advances in targeted therapy have focused on tumor necrosis factor-alpha (TNF-) inhibition, given its significant role in radicular symptoms. Freeman et al. conducted a multicenter, randomized, double-blind, placebo-controlled phase IIa clinical trial examining etanercept, a TNF- inhibitor, injected transforaminally for symptomatic disc herniation.63 Compared to placebo, etanercept proved both safe and effective in providing significant therapeutic relief for patients with sciatica, with efficacy similar to epidural dexamethasone injection.64

Surgical Outcomes and Comparative Effectiveness

The comparison between operative and nonoperative treatment has been extensively studied, with several landmark investigations providing crucial insights. The Maine Lumbar Spine Study group published comprehensive long-term results comparing surgically and nonsurgically treated patients over multiple time points.1,67 At 1-year follow-up, surgically treated patients demonstrated superior outcomes, with 71% reporting relief of back or leg pain compared to 43% of the nonoperative group, despite the surgical group being more symptomatic at initial presentation.

The durability of surgical benefits was confirmed in 5-year follow-up data, where 70% of surgical patients reported back or leg pain improvement compared to 56% of nonoperatively treated patients.1 However, the study revealed that surgical benefits were most pronounced in the early postoperative period, within the first 2 years, with advantages becoming less apparent at final follow-up. The 10-year results maintained the superiority of surgical treatment, with a significantly larger percentage of surgical patients reporting relief of low back and leg pain, better function, and higher satisfaction compared with nonoperative patients.7

The landmark SPORT investigation, designed as a rigorous randomized prospective controlled study, faced challenges with high crossover rates between treatment groups.8,9,68 Despite these methodological difficulties, both randomized and observational arms demonstrated improvement in bodily pain, physical function, and Oswestry Disability Index scores regardless of intervention. In as-treated analysis, surgical treatment showed statistically superior results compared with nonoperative treatment, with these benefits proving durable through 8-year follow-up reports.69

Weber's classic randomized prospective study provided additional long-term perspective on surgical versus nonsurgical outcomes.10 At 1 year, good results were reported in 33% of the nonoperative group versus 66% in the surgical group. However, outcomes converged over time, with results at 10 years showing 55% versus 57% good outcomes for nonsurgical and surgical groups, respectively. This convergence pattern suggests that while surgery provides earlier symptomatic relief, long-term outcomes may be similar between treatment approaches.

Prognostic Factors and Patient Selection

Several factors influence outcomes in lumbar disc disease, with timing of intervention being particularly crucial for surgical candidates. Rotheorl and colleagues demonstrated that patients with symptom duration exceeding 2 months had statistically significantly worse outcomes than those operated within 2 months.33 Similarly, Hurme and Alaranta found optimal results in patients operated within 2 months of disabling sciatica onset.12 Nygaard and colleagues reported inferior results in patients with leg pain duration of 8 months or more, while Jansson and colleagues confirmed similar findings in their analysis of Swedish registry data including over 27,000 patients.32,36

Patient selection criteria significantly impact outcomes, with ideal candidates demonstrating concordant clinical and radiological findings. Patients presenting with clear neurological deficits, positive imaging findings, and strong motivation for return to normal activities typically achieve the best results. Conversely, involvement in litigation, disability claims, or workers' compensation issues may negatively impact outcomes, as may underlying psychological factors.

Alaranta and colleagues provided important insights into patient subgroups by comparing operated versus nonoperated patients stratified by myelographic findings.70 At 1-year follow-up, 91% of operated patients and 82% of nonoperatively treated patients with positive myelograms showed improved pain levels. However, only 51% of nonoperatively treated patients with negative myelograms improved, suggesting a distinctly worse natural history for patients with sciatica but no evidence of root compression.

Complications and Revision Surgery

Surgical outcomes must be considered in the context of potential complications and the need for revision procedures. The Maine Lumbar Spine Study reported reoperation rates of 20% in operative patients, while 16% of patients initially treated nonsurgically eventually underwent operation.1 These figures highlight the dynamic nature of treatment decisions and the potential for treatment crossover over time.

Emerging technologies, including annular repair techniques following lumbar discectomy, represent attempts to improve outcomes and reduce revision rates. While one published prospective randomized trial of annular repair did not demonstrate statistically significant benefit in patient symptoms, it did reduce the need for subsequent surgery.99

Long-term Functional Outcomes

Long-term functional outcomes demonstrate generally favorable results for both surgical and nonsurgical treatment approaches. Work and disability status showed similar patterns between treatment groups in long-term follow-up studies, suggesting that while surgery may provide faster symptom relief, ultimate functional capacity may be comparable between approaches.7 For workers' compensation patients specifically, no difference in time to return to work was observed between operative and nonoperative groups, indicating that compensation status may influence outcomes regardless of treatment modality.

The prognosis for lumbar disc disease remains generally favorable, with most patients achieving meaningful improvement through either conservative or surgical management. The choice between treatment approaches should be individualized based on patient factors, symptom severity, functional limitations, and patient preferences, with the understanding that surgical intervention typically provides faster relief while long-term outcomes may converge between treatment modalities. Early intervention, within appropriate timeframes, appears to optimize outcomes regardless of the chosen treatment approach.

6 Conclusions

[Content pending for Conclusions]

References

1. Unknown (n.d.). *Entire Books.*